

☆ Number of Binary Trees using N nodes

For Unlabelled
Nodes,

$$T(n) = \frac{2^n C_n}{n+1}$$

→ Catalan
Number

$$\text{Max}^m \text{ height} = 2^{n-1}$$

Trees

Combination
Formulae

$$T(n) = \sum_{i=1}^n T(i-1) \times T(n-i)$$

→ Recursive
Formulae

For Labelled Nodes,

$$T(n) = \frac{2^n C_n}{n+1} \times n! \rightarrow \text{for Filling of values}$$

for shapes
like earlier